

Telecommunications Customer Churn Analysis

Final Project Report

1. Executive Summary

This report presents an Excel-based analysis of customer churn in a telecommunications company. The objective was to identify the major factors associated with customer churn, estimate its business impact, and communicate the findings through an interactive dashboard.

2. Project Objectives

- Analyze customer churn patterns.
- Identify key churn indicators.
- Measure the financial impact of churn.
- Develop an interactive Excel dashboard for business users.

3. Dataset

The analysis was performed using the provided telecommunications datasets:

- train.csv
- test.csv
- sampleSubmission.csv

4. Tools and Techniques

- Microsoft Excel
- Pivot Tables and Pivot Charts
- Excel Functions and Helper Columns
- Quartile Analysis
- Threshold Analysis
- Segmentation Analysis
- Interactive Dashboard with Slicers

5. Analysis Summary

Basic, medium, and advanced analyses were performed to evaluate customer behavior, service usage, subscription plans, customer support interactions, and estimated revenue impact.

6. Key Findings

- Overall churn rate: 14.07%.
- Customers making four or more customer service calls exhibited significantly higher churn.
- Customers in the highest day usage quartile showed the highest churn rate.
- International Plan subscribers experienced substantially higher churn than non-subscribers.
- Estimated monthly revenue loss due to churn: approximately ■35.6K.

7. Dashboard

An interactive Excel dashboard was developed featuring:

- KPI cards (Total Customers, Churn Rate, Churned Customers, Monthly Revenue Loss)
- Churn Rate by Customer Service Calls
- Churn Rate by Day Usage Quartile
- Revenue Impact Analysis
- Interactive slicers for International Plan, Voice Mail Plan, and Area Code.

8. Business Recommendations

- Prioritize customers with repeated customer service interactions.
- Monitor heavy daytime users and offer appropriate plans.
- Review International Plan pricing and customer value.
- Use dashboard insights to support proactive retention strategies.

9. Conclusion

Customer behavior and service interaction patterns are stronger indicators of churn than demographic or geographic attributes. The completed dashboard provides an accessible decision-support tool for monitoring churn drivers and supporting retention initiatives.